TOPIC 1 • THEORY Advanced Mathematics • BSc Aerospace Engineering • UC3M

# Introduction, Modelling and the Theory of First-Order Initial Value Problems

What a differential equation is, how to classify it, where the equations of engineering come from, and when an initial value problem actually has one and only one solution. Sections 1 to 6 follow page 2 of the class notes item by item; Robinson formalises and extends each step, and material beyond the notes is labelled as a complement.

**Ground truth:** Robinson (2004) Ch. 3 (pp. 11–16), Sec. 5.3 (pp. 29–31), Ch. 6 (pp. 38–45), Sec. 7.5.2 (p. 52), Ch. 8 Sec. 8.2–8.5 (pp. 61–68) • Class notes Notes.pdf, p. 2 • Lecture note *Definition of linear ODE* (Sep 9) • Lecture map ODE_lecture_theory_to_Robinson_chapters.txt , Part I (Sep 8–10)

≈ 55 min study time • 70 displayed equations • 7 figures

SECTION 01 • NOTES P. 2 • ROBINSON SEC. 3.1 • LECTURE SEP 8

## 1. What a differential equation is

Physical laws are rarely statements about quantities themselves; they are statements about how quantities *change*. Newton's second law relates force to the rate of change of momentum, a mass balance relates the net inflow to the rate of change of the stored mass, and a population model relates the number of births per year to the current population. Writing any of these laws down produces an equation in which the unknown is a **function**, and that function appears together with some of its derivatives. Such an equation is a **differential equation**.

DEFINITION — DIFFERENTIAL EQUATION [NOTES, P. 2]

A **differential equation** is an equation involving an unknown function (of one or more independent variables) and its derivatives.

DEFINITION — INDEPENDENT AND DEPENDENT VARIABLES

The **independent variables** are the arguments of the unknown function (e.g. $t$ in $x(t)$, $x$ in $y(x)$, both $x$ and $y$ in $G(x, y)$). The unknown function itself is the **dependent variable**. A differential equation relates a dependent variable, its derivatives and the independent variables. [Robinson, Sec. 3.1, p. 11]

**Notation.** The class notes write every derivative with a prime, also in time: $x'(t) = dx/dt$. Robinson keeps the prime for derivatives with respect to $x$ and uses a dot for derivatives with respect to time:

$$\dot{z} = \frac{dz}{dt}, \qquad \ddot{z} = \frac{d^2 z}{dt^2}, \qquad y' = \frac{dy}{dx}, \qquad y'' = \frac{d^2 y}{dx^2}.$$

Both notations appear on this page; they mean the same thing.

### 1.1 Ordinary versus partial differential equations

The most important distinction depends on how many independent variables there are. With one independent variable only ordinary derivatives can occur; with two or more, the derivatives are partial.

ODE AND PDE, WITH THE EXAMPLES OF THE CLASS NOTES [NOTES, P. 2]

- An **ODE** is a differential equation in which the unknown function depends on **only one** independent variable. Example: $x'(t) = x(t)$, with dependent variable $x$ and independent variable $t$. The function $x(t) = e^t$ is a solution, since $\frac{d}{dt}e^t = e^t$.
- A **PDE** is a differential equation in which the unknown function depends on **more than one** variable. Example: the heat equation for $u(x, t)$,

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}.$$

(The notes write the derivatives with $d$; with two independent variables they are partial derivatives.)

Robinson gives the same definitions with a wider set of examples [Robinson, Sec. 3.1, pp. 11–12]:

| EQUATION | NAME / MEANING | TYPE | SOURCE |
|---|---|---|---|
| $\frac{dy}{dx} = f(x)$ | Prescribed slope of the graph of $y$ | ODE, independent variable $x$ | p. 11 |
| $m\,\ddot{\mathbf{x}} = \mathbf{f}(t)$ | Newton's second law for a particle at $\mathbf{x}(t) = (x, y, z)$ | ODE (system of 3), variable $t$ | p. 12 |
| $-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi = E\psi$ | Schrödinger equation, complex $\psi(x)$ | ODE, variable $x$ | p. 12 |
| $\frac{\partial h}{\partial t} = k\,\frac{\partial^2 h}{\partial x^2}$ | Heat equation for the temperature $h(x, t)$ of a rod | PDE, variables $x, t$ | p. 12 |
| $\rho\big(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u}\cdot\nabla)\mathbf{u}\big) - \mu\Delta\mathbf{u} + \nabla p = \mathbf{f},\ \ \nabla\cdot\mathbf{u} = 0$ | Navier–Stokes equations for a fluid velocity $\mathbf{u}(x_1, x_2, x_3, t)$ | PDE system, 4 variables | Eqs. (3.1)–(3.3), p. 12 |

AEROSPACE RELEVANCE

The Navier–Stokes equations (3.3) are the model behind every computational fluid dynamics code used to design wings, intakes and nozzles. They are PDEs; this course deals with ODEs only [Robinson, p. 12]. ODEs still appear constantly in aerospace work: flight-path and attitude dynamics ($t$ is the only independent variable), structural vibration modes, and the self-similar reductions of the flow equations, such as the boundary-layer equation of Section 2.

## 2. Order and autonomy

**DEFINITION — ORDER** [NOTES, P. 2]

The **order** of an ODE is the order of the highest derivative in the ODE.

Robinson makes this precise [Robinson, Sec. 3.2, Eq. (3.4), p. 13]: an $n$th-order ODE for $y(t)$ is an equation of the form

$$F\left(\frac{d^n y}{dt^n},\ \frac{d^{n-1}y}{dt^{n-1}},\ \ldots,\ \frac{dy}{dt},\ y,\ t\right) = 0, \tag{3.4}$$

in which $d^n y/dt^n$ really occurs in $F$. The last condition is not pedantry. If $F(\ddot{y}, \dot{y}, y, t) = y - t$, then "$F = 0$" is the algebraic relation $y = t$, which contains no derivative at all and so is not a differential equation [p. 13]. The examples of the class notes: $x' = x$ is first order, the heat equation is second order (in $x$), and Newton's law $m\,y'' = F$ (Section 6) is second order.

| EQUATION | PHYSICAL MEANING | ORDER |
|---|---|---|
| $\frac{dy}{dx} = f(x)$ | Slope of a graph | 1 |
| $\frac{1}{2}m\dot{x}^2 + V(x) = E$ | Energy conservation: kinetic plus potential energy is constant | 1 |
| $m\frac{d^2x}{dt^2} = F$ | Newton's second law | 2 |
| $\psi''' + \frac{1}{2}\psi\psi'' = 0$ | Fluid boundary layers (known in fluid mechanics as the Blasius equation for flow along a flat plate) | 3 |

[Robinson, Sec. 3.2, p. 13]

**DEFINITION — AUTONOMOUS EQUATION** [ROBINSON, P. 13]

If the independent variable $t$ does not occur explicitly in the equation, as in $\dfrac{dy}{dt} = f(y)$, the equation is **autonomous**. Otherwise it is **non-autonomous**.

In the class notes autonomy appears on p. 3, where it is used for the stability of first-order equations (Topic 2). Its physical meaning is that the law does not depend on the clock. If $y(t)$ solves $\dot{y} = f(y)$, then so does the time-shifted function $z(t) = y(t - s)$ for any fixed $s$. To see this, apply the chain rule with the inner function $u(t) = t - s$, $du/dt = 1$:

$$\frac{dz}{dt} = \frac{dy}{du}\Big|_{u=t-s} \cdot \frac{du}{dt} = f\big(y(t-s)\big)\cdot 1 = f\big(z(t)\big).$$

So an experiment started today and the same experiment started tomorrow evolve identically. This is what makes the geometric phase-line method of Topic 2 work.

**HIGHER ORDER MEANS MORE STATE, NOT MORE DIFFICULTY IN PRINCIPLE**

Any second-order equation $\ddot{x} = f(\dot{x}, x, t)$ is equivalent to a first-order system for the two unknowns $x_1 = x, x_2 = \dot{x}$:

$$\dot{x}_1 = x_2, \qquad \dot{x}_2 = f(x_2, x_1, t).$$

The first equation is the definition of $x_2$; the second is the original equation rewritten. The same trick ( $x_k = y^{(k-1)}, k = 1, \dots, n$) turns any $n$th-order equation into $n$ first-order ones. This is developed in Topic 4 [Notes, p. 9; Robinson, Sec. 25.1, p. 251 for $n = 2$; Exercise 25.1, p. 252 for general $n$].

SECTION 03 • NOTES P. 2 • ROBINSON SEC. 3.3 • LECTURE SEP 8–10 (CORRECTED VERSION)

# 3. Linear and nonlinear equations

Linearity is the classification with the largest practical consequences. Robinson's summary is blunt: linear problems are relatively "easy" (an explicit solution can usually be found) and nonlinear problems are "hard" (explicit solutions exist only in very particular cases) [p. 13]. The class notes give two things: the form of a linear equation (in the margin of p. 2) and a test to decide linearity, based on superposition.

## 3.1 The form of a linear ODE

**LINEAR ODE** [NOTES, P. 2]

An ODE of order $n$ is **linear** if it has the form

$$a_0(x)\,y^{(n)}(x) + a_1(x)\,y^{(n-1)}(x) + \cdots + a_{n-1}(x)\,y'(x) + a_n(x)\,y(x) = b(x), \qquad \text{(L)}$$

where the coefficients $a_i$ and the right-hand side $b$ are functions of the independent variable only. It is **homogeneous** if $b \equiv 0$ and **inhomogeneous** otherwise.

The lecture note *Definition of linear ODE* (Sep 9) writes the same form with the indices in the opposite order, $a_n(x)\,y^{(n)} + \cdots + a_1(x)\,y' + a_0(x)\,y = b(x)$, and Robinson writes it with $t$ as the independent variable and $f(t)$ on the right, summarising it as "only multiples of $y$ and its derivatives occur"

[Robinson, Eq. (3.5), p. 14]. The coefficients may be as nonlinear as they like *in the independent variable*; what matters is how the *unknown* appears. Collecting the left-hand side of (L) into an operator,

$$L[y] = a_0\,y^{(n)} + a_1\,y^{(n-1)} + \cdots + a_n\,y = \sum_{j=0}^{n} a_j\,y^{(n-j)},$$

a linear ODE is simply $L[y] = b$.

## 3.2 The superposition test of the class notes

**LINEAR / NON-LINEAR ODES** [NOTES, P. 2]

Consider $u_1$ and $u_2$, two solutions to an ODE. Then the ODE is **linear** if, for all $\alpha, \beta \in \mathbb{R}$, $\alpha u_1 + \beta u_2$ is also a solution to the ODE. If not, the ODE is **non-linear**.

**SCOPE OF THE TEST**

"Every combination $\alpha u_1 + \beta u_2$ of solutions is a solution" is exactly the **superposition principle** of *homogeneous* linear equations. An inhomogeneous linear equation such as $x' - x = 1$ has the form (L) but fails the test as stated: with $\alpha = \beta = 0$ the zero function would have to be a solution, and $0 - 0 \neq 1$. For $b \not\equiv 0$ the test is applied to the homogeneous part $L[y] = 0$. The two examples of the notes below are tested exactly as written there.

## 3.3 Demonstration 1: a homogeneous linear equation passes the test

**PROPOSITION — SUPERPOSITION FOR $L[y] = 0$**

If $L[u_1] = 0$ and $L[u_2] = 0$, then $L[\alpha u_1 + \beta u_2] = 0$ for all $\alpha, \beta \in \mathbb{R}$.

**PROOF**

Set $v = \alpha u_1 + \beta u_2$.

1. Differentiation is linear: for every $k$, $v^{(k)} = \dfrac{d^k}{dx^k}(\alpha u_1 + \beta u_2) = \alpha u_1^{(k)} + \beta u_2^{(k)}$.

2. Substitute into $L$ and split the sum:

$$L[v] = \sum_{j=0}^{n} a_j\big(\alpha u_1^{(n-j)} + \beta u_2^{(n-j)}\big) = \alpha \sum_{j=0}^{n} a_j u_1^{(n-j)} + \beta \sum_{j=0}^{n} a_j u_2^{(n-j)} = \alpha L[u_1] + \beta L[u_2].$$

3. With $L[u_1] = L[u_2] = 0$: $L[v] = \alpha \cdot 0 + \beta \cdot 0 = 0$. $\square$

■

**EXAMPLE OF THE NOTES: IS $u' = u$ LINEAR?** [NOTES, P. 2]

Let $u_1$ and $u_2$ be solutions, $u_1' = u_1$ and $u_2' = u_2$, and call $u = \alpha u_1 + \beta u_2$. Then

$$u' = \alpha u_1' + \beta u_2' = \alpha u_1 + \beta u_2 = u,$$

so $u$ is again a solution for every $\alpha$, $\beta$: the ODE is **linear**. In the form (L): $n = 1, a_0 = 1, a_1 = -1, b = 0$.

The same property gives the structure of the solutions of an inhomogeneous equation: if $L[y_p] = b$ and $L[y_h] = 0$, then $L[y_h + y_p] = L[y_h] + L[y_p] = 0 + b = b$. General solution = homogeneous solution + one particular solution (Topic 3; Robinson Sec. 14.1).

### 3.4 Demonstration 2: $u'' = (u')^2 + 1$ fails the test

Let $u_1$ and $u_2$ be solutions, $u_1'' = (u_1')^2 + 1$ and $u_2'' = (u_2')^2 + 1$, and let $v = \alpha u_1 + \beta u_2$ [Notes, p. 2]. By linearity of differentiation,

$$v' = \alpha u_1' + \beta u_2', \qquad v'' = \alpha u_1'' + \beta u_2''.$$

For $v$ to be a solution we would need $v'' = (v')^2 + 1$. Compute both sides.

1. Left side, replacing $u_1''$ and $u_2''$ by their equations:

$$v'' = \alpha\big((u_1')^2 + 1\big) + \beta\big((u_2')^2 + 1\big) = \alpha\,(u_1')^2 + \beta\,(u_2')^2 + \alpha + \beta.$$

2. Right side, expanding the square $(a+b)^2 = a^2 + 2ab + b^2$:

$$(v')^2 + 1 = \big(\alpha u_1' + \beta u_2'\big)^2 + 1 = \alpha^2 (u_1')^2 + 2\alpha\beta\, u_1' u_2' + \beta^2 (u_2')^2 + 1.$$

3. Compare. The left side has $\alpha$ where the right side has $\alpha^2$, $\beta$ where it has $\beta^2$, no cross term $2\alpha\beta\, u_1' u_2'$, and the constant $\alpha + \beta$ instead of 1. The equality cannot hold for all $\alpha$, $\beta$: the ODE is **non-linear**.

**EXTENSION — THE CULPRIT IS THE SQUARE, NOT THE "+1"**

For an inhomogeneous *linear* equation, combinations with $\alpha + \beta = 1$ are still solutions ($L[\alpha u_1 + \beta u_2] = (\alpha + \beta)\, b = b$). For $u'' = (u')^2 + 1$ even these fail. Put $\beta = 1 - \alpha$ and subtract the two sides of steps 1 and 2; since $\beta(1-\beta) = (1-\alpha)\alpha$ and $2\alpha\beta = 2\alpha(1-\alpha)$,

$$v'' - \big((v')^2 + 1\big) = \alpha(1-\alpha)(u_1')^2 + \beta(1-\beta)(u_2')^2 - 2\alpha\beta\, u_1' u_2' = \alpha(1-\alpha)\big(u_1' - u_2'\big)^2,$$

which is not zero when $0 < \alpha < 1$ and $u_1' \neq u_2'$. Such solutions exist: $w = u'$ satisfies $w' = w^2 + 1$, solved by $w = \tan(t + c)$ because $\frac{d}{dt} \tan s = \sec^2 s = \tan^2 s + 1$, so $u = -\ln\cos(t + c) + d$. Take $u_1 = -\ln\cos t$, $u_2 = -\ln\cos(t - \pi/4)$ on $-\pi/4 < t < \pi/2$ and $\alpha = \beta = \frac{1}{2}$. At $t = 0$: $u_1' = 0$, $u_2' = -1$, $u_1'' = 1$, $u_2'' = 2$, so $v'' = \frac{3}{2}$ while $(v')^2 + 1 = \frac{1}{4} + 1 = \frac{5}{4}$. The gap $\frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} \cdot \big(0 - (-1)\big)^2$ is the one the formula predicts.

### 3.5 A practical test, and the lecture examples

Read each term of the equation and ask what it does to the unknown $y$. The equation is linear exactly when every term is one of:

1. a function of the independent variable alone (it goes into $b$);
2. a function of the independent variable multiplied by $y$ or by one derivative $y^{(k)}$, to the first power.

Anything else makes the equation nonlinear: powers such as $y^2$ or $(y')^2$, products such as $y\,y'$, and nonlinear functions of the unknown such as $\sin y$, $e^y$ or $\sqrt{y'}$.

The three equations analysed in class (corrected note dated Sep 9, presented in the lecture of Sep 10):

| ODE | ORDER | VERDICT | JUSTIFICATION AGAINST THE FORM (L) |
|---|---|---|---|
| $u'(t) = u(t)$ | 1 | **Linear**, homogeneous | Rewrite as $1 \cdot u' + (-1) \cdot u = 0$: $a_0 = 1$, $a_1 = -1$, $b = 0$. Superposition: Section 3.3. |
| $u''(t) = \big(u'(t)\big)^2 + 1$ | 2 | **Nonlinear** | The derivative $u'$ is squared. No choice of $a_i(t), b(t)$ reproduces $(u')^2$. Superposition fails: Section 3.4. |
| $y'(x) = 4\big(y(x)\big)^2 + x^2$ | 1 | **Nonlinear** | The unknown $y$ itself is squared. The term $x^2$ is harmless: it is a function of $x$ only and would sit in $b(x)$. |

LECTURE VS. BOOK: WHY A "CORRECTED VERSION"

The lecture of Sep 8 classified these examples and the solution was flagged as needing correction; the corrected solution was given on Sep 10 together with the note *Definition of linear ODE*. The table above follows the corrected note. The third example is a useful trap: a term like $x^2$ looks "nonlinear", but nonlinearity in the independent variable is irrelevant; only nonlinearity in the unknown counts.

### 3.6 Complement (Robinson): what "solving" can mean

Robinson lists the kinds of answer one can expect, in decreasing order of explicitness [Robinson, Sec. 3.4, pp. 14-16]:

| TYPE | EXAMPLE | COMMENT |
|---|---|---|
| Fully explicit | $y(t) = 3\cos 5t + 8\sin t$ (3.6) | Combination of elementary functions; rare. |
| Explicit with an integral | $y(t) = 1 + \int_0^t e^{-s^2}\,ds$ (3.7) | Still $y$ as a function of $t$, but the integral has no elementary antiderivative. |
| Implicit | $\ln y + 4\ln x - y - 2x + 4 = 0$ (3.8) | Relates $x$ and $y$ without derivatives; cannot be solved for $y$. |
| Qualitative | Phase portrait of (3.9), Fig. 1.1 | Describes the behaviour without any formula. |

| TYPE | EXAMPLE | COMMENT |
|---|---|---|
| Numerical | MATLAB `ode45` for $\dot{x} = t - x^2$, $x(0) = 0$ | A table of approximate values at discrete times. |

SECTION 04 • NOTES P. 2 • ROBINSON EQ. (3.9), CH. 25 • LECTURE SEP 8

# 4. Systems of ODEs

SYSTEMS OF ODES [NOTES, P. 2]

A **system of ODEs** has two or more dependent variables. Example:

$$\begin{cases} x'(t) = 2x(t) - y(t), \\ y'(t) = -x(t) + 3y(t). \end{cases}$$

The two unknowns $x(t), y(t)$ share the independent variable $t$. The system is **first order**, **linear** and homogeneous (each right-hand side is a combination of $x$ and $y$ with constant coefficients and nothing else), and autonomous. In matrix form,

$$\begin{pmatrix} x \\ y \end{pmatrix}' = \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

It passes the superposition test of Section 3.2: if $(x_1, y_1)$ and $(x_2, y_2)$ are solutions, then $(x, y) = \alpha(x_1, y_1) + \beta(x_2, y_2)$ satisfies $x' = \alpha x_1' + \beta x_2' = \alpha(2x_1 - y_1) + \beta(2x_2 - y_2) = 2x - y$, and in the same way $y' = -x + 3y$.

***Preview of Topic 4: one second-order equation for $x$***

Solve the first equation for $y$: $y = 2x - x'$. Differentiating, $y' = 2x' - x''$. Substitute both into the second equation:

$$2x' - x'' = -x + 3\,(2x - x') = 5x - 3x' \quad \Longrightarrow \quad x'' - 5x' + 5x = 0.$$

This is the elimination method of the class notes, p. 9, which turns $x' = ax + by$, $y' = cx + dy$ into $x'' - (a + d)x' + (ad - bc)x = 0$. Check: $a = 2, b = -1, c = -1, d = 3$ give $a + d = 5$ and $ad - bc = 6 - 1 = 5$. Second-order equations of this kind are solved in Topic 3 and systems in Topic 4.

### 4.1 Complement (Robinson): a nonlinear system and its phase portrait

Robinson's first example (which he interprets as two competing species only much later, p. 326) has unknowns $x(t)$ and $y(t)$ [Robinson, Eq. (3.9), p. 15]:

$$\dot{x} = x\,(4 - 2x - y), \qquad \dot{y} = y\,(9 - 3x - 3y). \tag{3.9}$$

Each species grows when rare (the constant terms $4x$ and $9y$), suffers from crowding by its own kind ($-2x^2$, $-3y^2$) and from competition with the other ($-xy$, $-3xy$). The system is first order, autonomous and nonlinear (it contains $x^2$, $y^2$ and $xy$), so the superposition test fails.

***Where the solutions settle***

A point where both right-hand sides vanish is a constant (equilibrium) solution. From $x(4 - 2x - y) = 0$ we need $x = 0$ or $y = 4 - 2x$; from $y(9 - 3x - 3y) = 0$ we need $y = 0$ or $y = 3 - x$. Combining the four options:

- $x = 0,\ y = 0$: the point $(0, 0)$;
- $x = 0,\ y = 3 - x$: the point $(0, 3)$;
- $y = 4 - 2x,\ y = 0$: the point $(2, 0)$;
- $y = 4 - 2x,\ y = 3 - x$: subtracting, $0 = 1 - x$, so $x = 1$ and $y = 2$: the point $(1, 2)$.

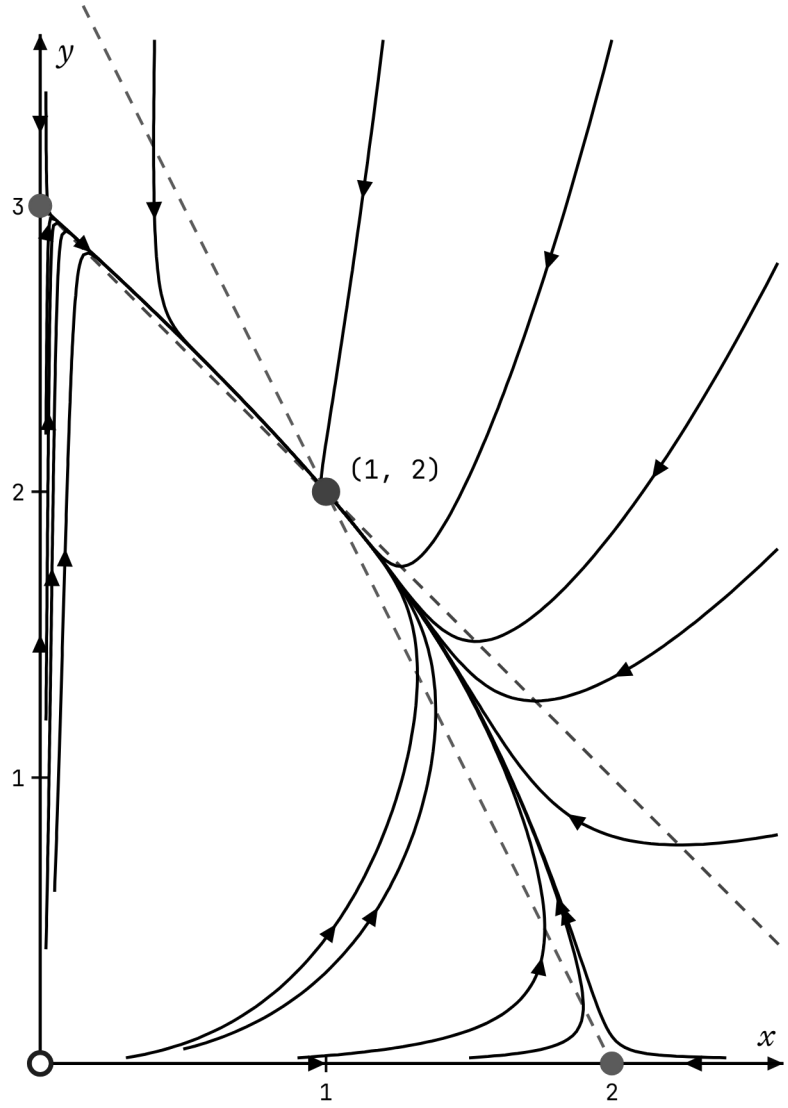



***Figure 1.1.*** *Phase portrait of system (3.9), $\dot{x} = x(4 - 2x - y)$, $\dot{y} = y(9 - 3x - 3y)$, drawn by numerical integration (after Robinson Fig. 3.1, p. 15). Dashed lines: curves where $\dot{x} = 0$ (amber) and $\dot{y} = 0$ (cyan). Every solution starting with $x > 0$, $y > 0$ tends to the green point $(1, 2)$, although no explicit formula for $x(t)$, $y(t)$ is known.*

The figure, which Robinson calls a **phase diagram** or **phase portrait**, shows that every solution with both species present tends to $(1, 2)$ as $t \to +\infty$: the species coexist `[p. 15; species reading, p. 326]`. The vector notation $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t)$ and the full theory of systems are the subject of Topic 4.

# 5. Example 1: population growth and decay

## 5.1 The model

**POPULATION GROWTH / DECAY** [NOTES, P. 2]

$x(t)$ is the population at time $t$, with $x(t) > 0$ for all $t$. The rate of change of $x$ is proportional to $x$ itself, with proportionality constant $k \in \mathbb{R}$. This gives the first-order linear ODE

$$x'(t) = k\,x(t), \qquad x(0) = x_0 > 0.$$

**Where the law comes from.** Robinson derives it from a balance [Robinson, Sec. 8.3, Eq. (8.5), p. 62]: in a short time $\Delta t$ a population $x$ gains about $\beta x\,\Delta t$ births and loses about $\delta x\,\Delta t$ deaths, with constant per-capita rates $\beta, \delta$. Hence $\Delta x \approx (\beta - \delta)\,x\,\Delta t$; dividing by $\Delta t$ and letting $\Delta t \to 0$ gives $x' = kx$ with $k = \beta - \delta$ (Malthus, 1798).

| SYMBOL | MEANING | SI UNIT |
|---|---|---|
| $x(t)$ | Population (number of individuals) | dimensionless count |
| $x_0$ | Initial population, $x(0)$ | dimensionless count |
| $k$ | Net growth rate per individual | $\mathrm{s}^{-1}$ (in practice $\mathrm{yr}^{-1}$) |
| $t$ | Time | s |

Linearity: in the form (L) with $t$ as variable, $1 \cdot x' + (-k) \cdot x = 0$, so $a_0 = 1, a_1 = -k, b = 0$.

## 5.2 Solution by integration from $0$ to $t$

Because $x(\tau) > 0$ we may divide by it: $\dfrac{x'(\tau)}{x(\tau)} = k$. The class notes integrate both sides from $0$ to $t$ (the dummy variable is $\tau$, so that $t$ can be the upper limit) and evaluate the left side in two equivalent ways [Notes, p. 2].

1. **Route 1: substitution.** Put $u = x(\tau)$, so $du = x'(\tau)\,d\tau$. The limits change with it: $\tau = 0$ gives $u = x(0) = x_0$ and $\tau = t$ gives $u = x(t)$. An antiderivative of $1/u$ for $u > 0$ is $\ln u$, so by Barrow's rule

$$\int_0^t \frac{x'(\tau)}{x(\tau)}\,d\tau = \int_{x_0}^{x(t)} \frac{du}{u} = \Big[\ln u\Big]_{x_0}^{x(t)} = \ln x(t) - \ln x_0 = \ln \frac{x(t)}{x_0}.$$

The notes abbreviate this step as $\dfrac{dx}{dt} = kx \Rightarrow \displaystyle\int_0^t \frac{dx}{x} = \int_0^t k\,dt.$

2 **Route 2: recognise a derivative.** By the chain rule with outer function ln and inner function $x(\tau) > 0$,

$$\frac{d}{d\tau} \ln\big(x(\tau)\big) = \frac{1}{x(\tau)} \cdot x'(\tau),$$

so the integrand is the derivative of $\ln x(\tau)$, and Barrow's rule gives directly

$$\int_0^t \big(\ln x(\tau)\big)' \, d\tau = \ln x(t) - \ln x(0) = \ln \frac{x(t)}{x_0}.$$

3 **Right side.** An antiderivative of the constant $k$ is $k\tau$:

$$\int_0^t k \, d\tau = \Big[k\tau\Big]_0^t = kt - k \cdot 0 = kt.$$

4 **Equate and solve.** $\ln \dfrac{x(t)}{x_0} = kt$. Exponentiate both sides: $\dfrac{x(t)}{x_0} = e^{kt}$.

SOLUTION

$$x(t) = x_0 \, e^{kt}.$$

- **Check the ODE:** $x'(t) = x_0 \, k \, e^{kt} = k \, x(t)$ (chain rule on $e^{kt}$).
- **Check the initial value:** $x(0) = x_0 e^0 = x_0$ .
- **Dimensions:** $[kt] = \mathrm{s}^{-1} \cdot \mathrm{s} = 1$, so the exponential is dimensionless and $x$ has the units of $x_0$ .

### 5.3 Growth or decay: the sign of $k$

| CASE [NOTES, P. 2] | BEHAVIOUR | AS $t \to \infty$ | TIME SCALE |
|---|---|---|---|
| $k > 0$ | Population grows | $x(t) \to \infty$ | Doubling time $d$: $2 = e^{kd}$, so $d = \ln 2/k$ |
| $k < 0$ | Population decays | $x(t) \to 0$ | Half-life: $\frac{1}{2} = e^{kh}$, so $h = \ln 2/\lvert k\rvert$ |
| $k = 0$ | No change | $x(t) \equiv x_0$ | – |

Neither time scale depends on $x_0$: a population of a thousand and one of a million double in the same time. The decaying case is also the law of radioactive decay (Problem 1.3, plutonium-239).

**COMPLEMENT (ROBINSON, SEC. 8.2): ANY STARTING TIME, ANY SIGN**

The notes assume $x > 0$, natural for a population. For $\dot{x} = \lambda x$ with $x(t_0) = x_0$ of either sign, Robinson argues that $x \equiv 0$ is the only solution through $x_0 = 0$ (uniqueness, Section 8), so no other solution can reach 0; $x(t)$ keeps the sign of $x_0$, the moduli in $\ln|x|$ can be dropped, and the same computation gives $x(t) = x_0\, e^{\lambda(t-t_0)}$ [Robinson, Sec. 8.2, pp. 61–62].

**COMPLEMENT: CENSUS TEST OF THE MODEL (ROBINSON SEC. 8.3, PP. 63–64)**

Population of Great Britain and Ireland: 1801: 16 345 646; 1851: 27 533 755; 1901: 41 609 091 (Eq. 8.6).

Fitting $k$ from 1801 and 1851, $x(1851) = x(1801)e^{50k}$:

$$k = \frac{\ln x(1851) - \ln x(1801)}{50} = \frac{\ln(27\,533\,755/16\,345\,646)}{50} = \frac{0.52145}{50} = 0.01043\ \text{yr}^{-1}.$$

- Prediction for 1901: $x(1801)e^{100k} = 46.4$ million (true: 41.6 million) – reasonable.
- Prediction for 2001: $x(1801)e^{200k} = 131.6$ million (true: about 63 million, UK plus Republic of Ireland) – more than twice too large.
- Doubling time: $\ln 2/0.01043 = 66.5$ years. Robinson rounds $k$ to 0.010 first and quotes 69 years; both describe the same fit.

## 5.4 The logistic correction

**WHY THE MODEL MUST BE CORRECTED** [NOTES, P. 2]

In reality a population does not grow exponentially forever, but reaches a flat maximum: we use a **logistic correction**.

The logistic equation keeps the exponential rate for small populations but reduces it as the population approaches a maximum sustainable value $M$ [Robinson, Sec. 8.5, Eq. (8.12), p. 66]. In the notation of the notes, $x' = kx\,(1 - x/M) = kx - \frac{k}{M}x^2$. The class notes analyse it on p. 3, with the first-order methods: its stationary points and their stability, and its explicit solution by separation of variables (Topic 2). Robinson's treatment follows, in his notation $p$ for the population.

$$\frac{dp}{dt} = kp\left(1 - \frac{p}{M}\right), \qquad k > 0,\ M > 0. \tag{8.12}$$

- For $p \ll M$, $p^2/M$ is negligible and $\dot{p} \approx kp$: the exponential model is recovered, so $k$ is the growth rate of small populations.
- For $0 < p < M$, $\dot{p} > 0$: the population grows. For $p > M$, $\dot{p} < 0$: it shrinks.
- The equation is first order, autonomous and nonlinear (the term $-kp^2/M$).

The equilibria are the zeros of $kp(1-p/M)$: $p=0$ and $p=M$. The phase-line analysis of Topic 2 shows that $p=0$ is unstable and $p=M$ is stable, so every positive population eventually settles at $M$.

***Explicit solution (Robinson pp. 66–67), every step shown***

1. Separate the variables (valid while $p \neq 0, M$, which by uniqueness then holds for all $t$), multiplying top and bottom by $M$:

$$\frac{M}{p\,(M-p)}\,dp = k\,dt.$$

2. Partial fractions: we want $\dfrac{M}{p(M-p)} = \dfrac{\alpha}{p} + \dfrac{\beta}{M-p}$, i.e. $M = \alpha(M-p) + \beta p$ for all $p$. Setting $p=0$ gives $\alpha = 1$; setting $p=M$ gives $\beta = 1$. So (Eq. 8.13)

$$\left(\frac{1}{p} + \frac{1}{M-p}\right) dp = k\,dt.$$

3. Antiderivatives: $\int dp/p = \ln|p|$; for the second, substitute $u = M-p$, $du = -dp$, so $\int \frac{dp}{M-p} = -\int \frac{du}{u} = -\ln|M-p|$. Hence $\int \left(\frac{1}{p} + \frac{1}{M-p}\right) dp = \ln|p| - \ln|M-p|$.

4. Barrow's rule between $t_0$ and $t$:

$$\Big[\ln|p| - \ln|M-p|\Big]_{p(t_0)}^{p(t)} = \Big[k\tilde{t}\Big]_{t_0}^{t} \implies \ln\frac{p(t)\,|M-p(t_0)|}{|M-p(t)|\,p(t_0)} = k(t-t_0).$$

   ($p>0$, so $|p| = p$.)

5. $M-p(t)$ never changes sign (a solution cannot cross the equilibrium $p=M$), so the moduli cancel. Exponentiating:

$$\frac{p(t)\,\big(M-p(t_0)\big)}{\big(M-p(t)\big)\,p(t_0)} = e^{k(t-t_0)}.$$

6. Solve for $p(t)$. Write $E = e^{k(t-t_0)}$ and $p_0 = p(t_0)$: $p\,(M-p_0) = (M-p)\,p_0 E$, so $p\,(M-p_0+p_0E) = Mp_0E$ and

**LOGISTIC SOLUTION**

$$p(t) = M\,\frac{p_0\,e^{k(t-t_0)}}{M - p_0 + p_0\,e^{k(t-t_0)}}.$$

- **Check at $t = t_0$:** $E = 1, p = Mp_0/M = p_0$ .
- **Long time:** divide top and bottom by $E$: $p = Mp_0/\big((M - p_0)e^{-k(t-t_0)} + p_0\big) \to Mp_0/p_0 = M$ .
- **Small population:** if $p_0 E \ll M$, the denominator is $\approx M$ and $p \approx p_0 e^{k(t-t_0)}$ (exponential growth) .
- **Inflection:** $\ddot{p} = \frac{d}{dt}\left[kp - kp^2/M\right] = k\dot{p}\,(1 - 2p/M)$ by the chain rule, so the growth is fastest at $p = M/2$.

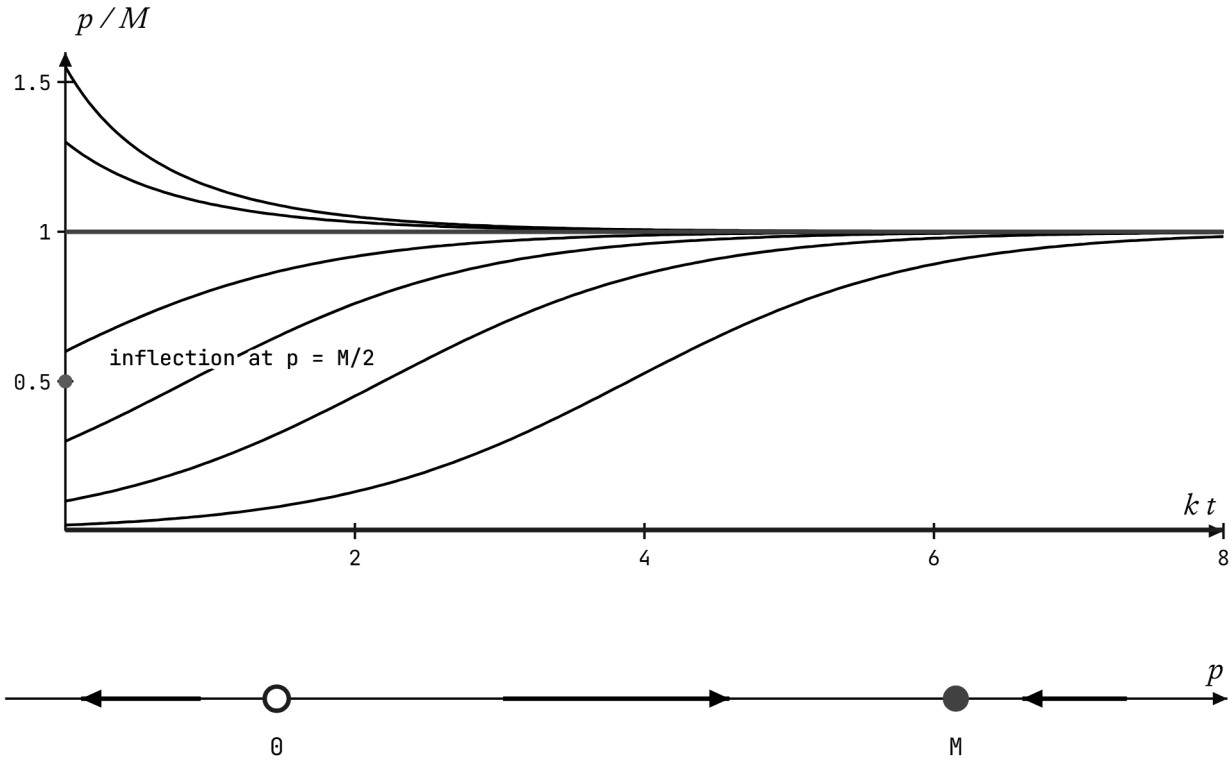



***Figure 1.4.*** *Top: solutions of the logistic equation (8.12) in units $p/M$ and $kt$, for several initial values. Bottom: the phase line (Robinson Fig. 8.3, p. 66), with the unstable equilibrium $p = 0$ (hollow) and the stable equilibrium $p = M$ (filled).*

CENSUS FIT OF THE LOGISTIC MODEL (ROBINSON PP. 67–68)

With $\alpha = e^{50k}$, $p_0 = p(1801)$, $p_1 = p(1851)$, $p_2 = p(1901)$, the solution requires $p_1 = \dfrac{Mp_0\alpha}{M - p_0 + p_0\alpha}$ and $p_2 = \dfrac{Mp_0\alpha^2}{M - p_0 + p_0\alpha^2}$. Eliminating $M$ gives

$$\alpha = \frac{p_2(p_1 - p_0)}{p_0(p_2 - p_1)} = 2.0234, \qquad M = \frac{p_1(2p_0p_2 - p_1p_2 - p_0p_1)}{p_0p_2 - p_1^2} = 83.1 \text{ million}, \tag{8.14}$$

so $k = \ln(2.0234)/50 = 0.0141\ \text{yr}^{-1}$. The model then predicts 66.8 million for 2001, against about 63 million observed – far better than the exponential model's 131.6 million (values recomputed from the census data; they agree with Robinson's).

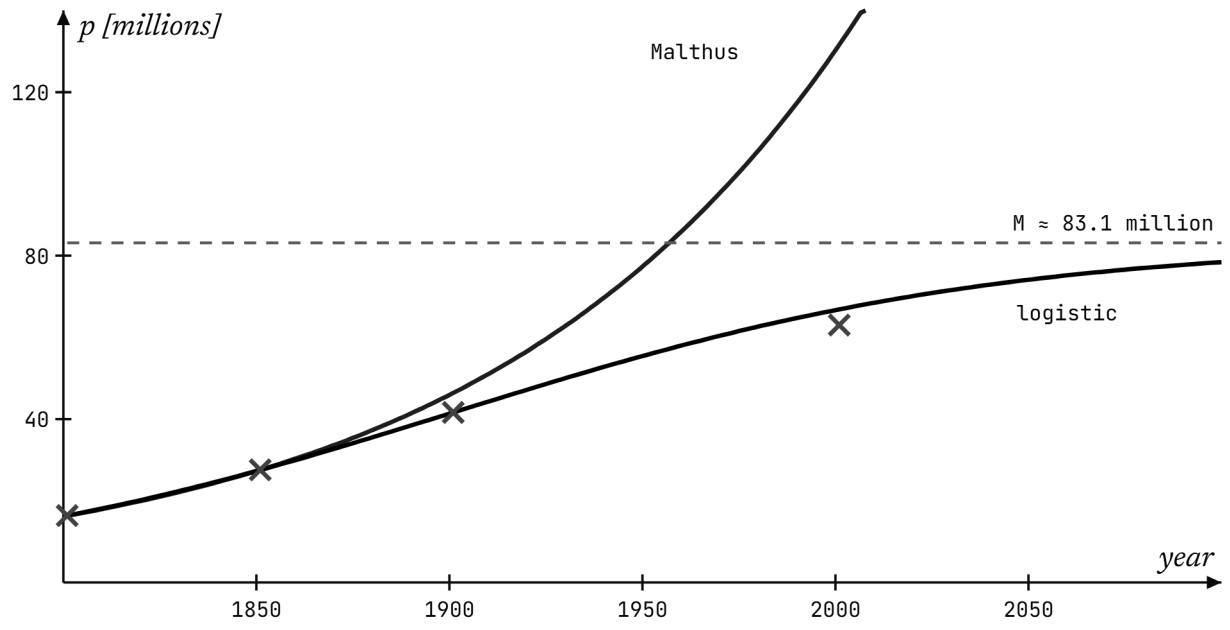



***Figure 1.3.*** *Census data for Great Britain and Ireland (green crosses: 1801, 1851, 1901 and ≈ 63 million in 2001) with the Malthus fit $k \approx 0.0104$ (red) and the logistic fit $k \approx 0.0141$, $M \approx 83.1$ million (indigo). Recomputed from the data of Robinson (8.6), p. 63; compare Figs. 8.2 and 8.4.*

SECTION 06 • NOTES P. 2 • ROBINSON SEC. 5.3, SEC. 7.5.2, EX. 8.6 • LECTURE SEP 8

# 6. Example 2: Newton's second law and free fall

## 6.1 The model

NEWTON'S 2ND LAW, "$F = ma$": FREE FALL OF AN OBJECT UNDER $g$ [NOTES, P. 2]

Let $y(t)$ be the height of the object, measured on a vertical axis pointing up. The only force is the weight, $F = -mg$, and the acceleration is $y''(t)$, so

$$-m\,g = m\,y''(t) \quad \Longrightarrow \quad \frac{d^2y}{dt^2} = -g,$$

a **second-order linear** ODE. The mass cancels.

**Hypotheses.** Constant gravity $g \approx 9.81\ \mathrm{m/s^2}$, no air resistance, motion along a vertical line: one degree of freedom, the height $y(t)$ [m]. In the form (L): $a_0 = 1, a_1 = a_2 = 0, b = -g$; the equation is inhomogeneous with constant right-hand side.

### 6.2 Integrating twice

The notes write the answer as $y = -\iint g\,dt\,dt$. With the antiderivatives and the limits shown:

1. Write $y'' = (y')'$. An antiderivative of the constant $-g$ is $-g\tau$, so integrating from $0$ to $t$ and using Barrow's rule,

$$y'(t) - y'(0) = \int_0^t (-g)\,d\tau = \Big[-g\tau\Big]_0^t = -gt.$$

   With $v_0 = y'(0)$: $y'(t) = v_0 - gt$.

2. An antiderivative of $v_0 - g\tau$ is $v_0\tau - \tfrac{1}{2}g\tau^2$ (check: $\frac{d}{d\tau}\big(v_0\tau - \tfrac{1}{2}g\tau^2\big) = v_0 - g\tau$), so

$$y(t) - y(0) = \int_0^t (v_0 - g\tau)\,d\tau = \Big[v_0\tau - \tfrac{1}{2}g\tau^2\Big]_0^t = v_0 t - \tfrac{1}{2}gt^2.$$

SOLUTION

$$y(t) = y_0 + v_0\,t - \tfrac{1}{2}g\,t^2, \qquad y_0 = y(0), \quad v_0 = y'(0).$$

Two integrations produce two constants, one per order of the equation. The notes take the object released from rest at the origin, $y_0 = v_0 = 0$:

$$y(t) = -\iint g\,dt\,dt = -\tfrac{1}{2}g\,t^2.$$

- **Check:** $y' = -gt, y'' = -g$ ; $y(0) = 0, y'(0) = 0$ .
- **Dimensions:** $[gt^2] = \mathrm{m\,s^{-2}} \cdot \mathrm{s^2} = \mathrm{m}$ .
- **Mass independence:** $m$ cancelled – without drag every body falls the same way.

### 6.3 Complement (Robinson): from momentum balance to $F = ma$

Newton's second law states that the change $\Delta p$ in the momentum of a body equals the applied force $F$ times the time $\Delta t$ over which it acts, $\Delta p = F\,\Delta t$ [Robinson, Sec. 5.3, p. 30]. Dividing by $\Delta t$ and letting $\Delta t \to 0$ (Robinson notes this step is somewhat imprecise) gives

$$\frac{dp}{dt} = F(t).$$

With $p = mv$, the product rule gives

$$\frac{dp}{dt} = \frac{d}{dt}(mv) = \frac{dm}{dt}\,v + m\,\frac{dv}{dt}.$$

For constant mass $dm/dt = 0$, so $m\,\dfrac{dv}{dt} = F(t)$: the familiar $F = ma$. (For variable-mass bodies such as rockets, $F = d(mv)/dt$ does not apply: the momentum carried away by the exhaust must be included in a balance over the whole system, which is outside this course.)

### 6.4 Complement (Robinson Exercise 5.6): release from a height $h$

Release the body from rest at height $h$: $y_0 = h$, $v_0 = 0$ in the solution of 6.2, so $y(t) = h - \frac{1}{2}gt^2$ and $y'(t) = -gt$. The book leaves this case as Exercise 5.6 (an apple falling from height $h$).

1. The impact time $t^*$ solves $y(t^*) = 0$: $h = \frac{1}{2}gt^{*2}$, so $t^* = \sqrt{2h/g}$ (the positive root, since $t^* > 0$).
2. The impact velocity is $y'(t^*) = -g\sqrt{2h/g} = -\sqrt{2gh}$.

RESULT AND CHECKS

$$y(t) = h - \tfrac{1}{2}gt^2, \qquad y'(t) = -gt, \qquad t^* = \sqrt{\frac{2h}{g}}, \qquad |y'(t^*)| = \sqrt{2gh}.$$

- **Dimensions:** $[\sqrt{h/g}] = \sqrt{\mathrm{m}/(\mathrm{m\ s^{-2}})} = \mathrm{s}$ ; $[\sqrt{gh}] = \sqrt{\mathrm{m^2 s^{-2}}} = \mathrm{m/s}$ .
- **Energy:** $\frac{1}{2}m\,y'(t^*)^2 = \frac{1}{2}m(2gh) = mgh$: the potential energy lost equals the kinetic energy gained .

EXAMPLE 5.4 (ROBINSON PP. 30–31; NUMBERS FROM THE CAPTION OF FIG. 5.6): A BRAKING CAR — THE SAME METHOD, HORIZONTALLY

A car of mass $m$ moving at $v_0$ brakes with a constant force $k$. Newton: $m\dot{v} = -k$. Integrating from 0 to $t$, $v(t) - v_0 = \int_0^t (-k/m)\,d\tilde{t} = -kt/m$, so $v(t) = v_0 - kt/m$. It stops when $v(t_c) = 0$, i.e. $t_c = mv_0/k$. Integrating $\dot{x} = v_0 - kt/m$ from 0 to $t_c$:

$$x(t_c) - x(0) = \Big[v_0\tilde{t} - \frac{k\tilde{t}^{\,2}}{2m}\Big]_0^{t_c} = v_0 t_c - \frac{kt_c^2}{2m} = \frac{mv_0^2}{k} - \frac{mv_0^2}{2k} = \frac{mv_0^2}{2k}.$$

With Robinson's data $m = 1000$ kg, $k = 6500$ N, $v_0 = 30$ mph $= 13.41$ m/s: $t_c = 1000 \times 13.41/6500 = 2.06$ s and the stopping distance is $1000 \times 13.41^2/(2 \times 6500) = 13.8$ m (about 45 ft, as in his Fig. 5.7). Dimensions: $[mv_0^2/k] = \mathrm{kg\ m^2 s^{-2}}/(\mathrm{kg\ m\ s^{-2}}) = \mathrm{m}$ .

### 6.5 Complement (Robinson): free fall with quadratic air drag, terminal velocity

At the speeds of a skydiver or a descending capsule, air resistance is not negligible. Robinson models it with a drag force proportional to the square of the speed and opposite to the motion [Sec. 7.5.2, Eq. (7.5), p. 52]:

$$m\frac{dv}{dt} = mg - kv|v|, \tag{7.5}$$

where now $v$ is the *downward* velocity and $k > 0$ (units kg/m) collects air density, drag coefficient and frontal area. The term $v|v|$ (rather than $v^2$) makes the drag oppose the motion whatever its direction. For a body released from rest, $v \geq 0$ and the equation becomes

$$\frac{dv}{dt} = g - \frac{k}{m}v^2.$$

This equation is first order, autonomous and **nonlinear** ($v^2$). Its right-hand side vanishes at $v^* = \sqrt{mg/k}$, the **terminal velocity**: if the body ever reaches $v^*$, weight and drag balance and it stays there. Robinson's skydiver ($m = 100$ kg, $k \approx 1/3$ kg/m, $g \approx 9.8$ m/s$^2$) has $v^* = \sqrt{100 \times 9.8 \times 3} \approx 54.2$ m/s [p. 53].

***Explicit solution (Robinson Exercise 8.6, worked here in full)***

Write $w_T = \sqrt{mg/k}$, so that $k/m = g/w_T^2$ and $\dot v = \dfrac{g}{w_T^2}\,(w_T^2 - v^2)$. For $0 \leq v < w_T$ we may divide (separation of variables, justified in Topic 2):

$$\frac{dv}{w_T^2 - v^2} = \frac{g}{w_T^2}\,dt.$$

1. Partial fractions: since $w_T^2 - v^2 = (w_T - v)(w_T + v)$,

$$\frac{1}{w_T^2 - v^2} = \frac{1}{2w_T}\left(\frac{1}{w_T - v} + \frac{1}{w_T + v}\right).$$

   Check: $\dfrac{(w_T + v) + (w_T - v)}{2w_T(w_T - v)(w_T + v)} = \dfrac{2w_T}{2w_T(w_T^2 - v^2)}$.

2. Antiderivatives. For $\int \frac{dv}{w_T - v}$ substitute $u = w_T - v$, $du = -dv$: $\int \frac{-du}{u} = -\ln|u| = -\ln(w_T - v)$ (here $w_T - v > 0$). Directly, $\int \frac{dv}{w_T + v} = \ln(w_T + v)$. Hence

$$\int \frac{dv}{w_T^2 - v^2} = \frac{1}{2w_T}\ln\frac{w_T + v}{w_T - v}.$$

3 Barrow's rule between $v(0)=0$ and $v(t)$ on the left, $0$ and $t$ on the right:

$$\frac{1}{2w_T}\left[\ln\frac{w_T+v}{w_T-v}\right]_0^{v(t)} = \frac{1}{2w_T}\left(\ln\frac{w_T+v(t)}{w_T-v(t)} - \ln 1\right) = \frac{g}{w_T^2}\left[\tilde{t}\right]_0^{t} = \frac{gt}{w_T^2}.$$

4 Multiply by $2w_T$ and exponentiate: $\dfrac{w_T+v}{w_T-v} = e^{2gt/w_T}$. Solve for $v$: $w_T+v = (w_T-v)e^{2gt/w_T}$, so $v\left(1+e^{2gt/w_T}\right) = w_T\left(e^{2gt/w_T}-1\right)$ and, dividing numerator and denominator by $e^{gt/w_T}$,

$$v(t) = w_T\,\frac{e^{gt/w_T}-e^{-gt/w_T}}{e^{gt/w_T}+e^{-gt/w_T}} = w_T\tanh\!\left(\frac{gt}{w_T}\right).$$

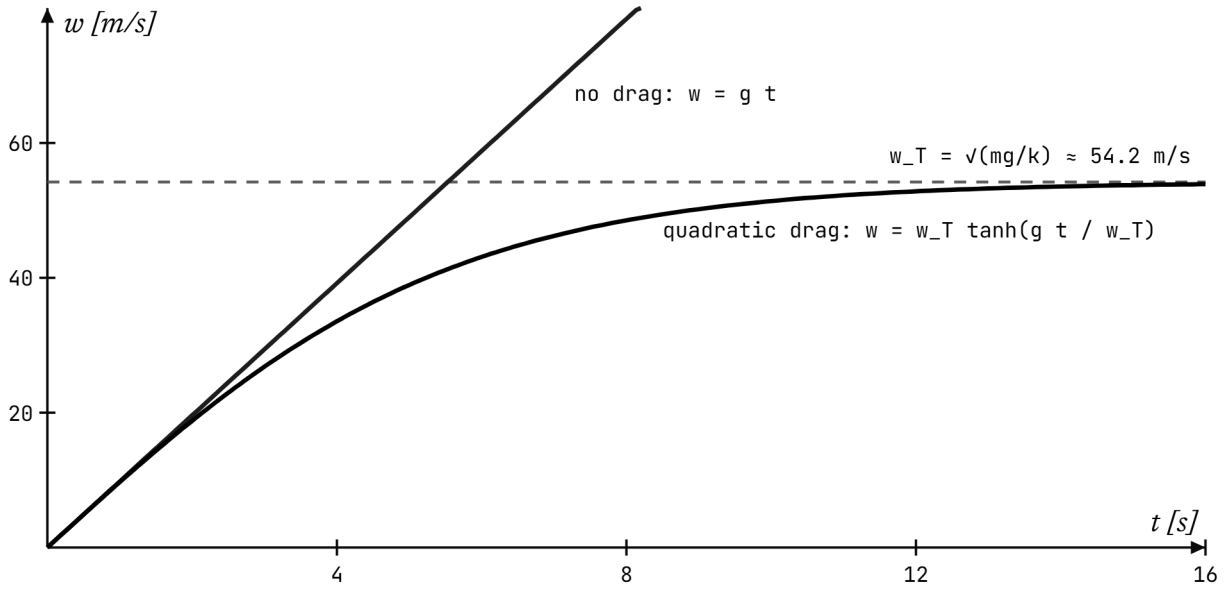



***Figure 1.2.*** *Downward speed of a skydiver released from rest, with Robinson's data $m = 100$ kg, $k \approx 1/3$ kg/m, $g \approx 9.8$ m/s$^2$ (Sec. 7.5.2, pp. 52–53). Without drag the speed grows linearly; with quadratic drag it saturates at the terminal speed.*

INTERPRETATION AND LIMITING CASES

- **Long times:** $\tanh s \to 1$ as $s \to \infty$, so $v(t) \to w_T$: the body approaches terminal velocity but never exceeds it.
- **Short times:** $\tanh s = s - s^3/3 + \dots$, so $v(t) \approx w_T \cdot gt/w_T = gt$: the drag-free result of 5.2 is recovered while $v \ll w_T$.
- **Vanishing drag:** $k \to 0$ gives $w_T \to \infty$ and again $v \to gt$ for every fixed $t$ (this is the limit Robinson asks for in Ex. 8.6).
- **Time scale:** $w_T/g$ is the time to reach about $76\%$ of $w_T$ ($\tanh 1 \approx 0.76$); for the skydiver $54.2/9.8 \approx 5.5$ s.

- **Dimensions:** $[kv^2] = \mathrm{kg\,m^{-1}} \cdot \mathrm{m^2 s^{-2}} = \mathrm{N}$ ; $[\sqrt{mg/k}] = \sqrt{\mathrm{kg\,m\,s^{-2}}/(\mathrm{kg\,m^{-1}})} = \mathrm{m/s}$ ; $[gt/w_T] = 1$ .

AEROSPACE CONNECTION

In aerodynamic notation the drag is $D = \tfrac{1}{2}\rho\, C_D A\, v^2$, so $k = \tfrac{1}{2}\rho C_D A$ and $w_T = \sqrt{2mg/(\rho C_D A)}$. Terminal velocity grows as the air thins ($\rho$ decreases with altitude), which is why a re-entering capsule or a high-altitude jumper falls much faster in the upper atmosphere. That refinement makes $k$ depend on height and is beyond this topic.

SECTION 07 • ROBINSON SEC. 6.1 • LECTURE SEP 10

# 7. Initial value problems and well-posedness

A differential equation alone has many solutions: $\dot{x} = \lambda x$ is solved by $Ae^{\lambda t}$ for every constant $A$. One extra piece of information, the state at one instant, selects one of them.

DEFINITION — INITIAL VALUE PROBLEM (IVP) [ROBINSON, SEC. 6.1, EQ. (6.2), P. 38]

An **initial value problem** is a differential equation together with an initial condition:

$$\frac{dx}{dt} = f(x,t), \qquad x(t_0) = x_0. \tag{6.2}$$

DEFINITION 6.1 — SOLUTION OF AN IVP [ROBINSON, P. 40]

Given an open interval $I$ that contains $t_0$, a **solution** of (6.2) on $I$ is a differentiable function $x(t)$ defined on $I$ with $x(t_0) = x_0$ and $\dot{x}(t) = f(x(t), t)$ for all $t \in I$.

The definition deliberately allows a solution to live on a limited interval $I$ rather than on all of $\mathbb{R}$. Section 9 shows that this is unavoidable even for very simple equations.

## 7.1 The philosophy of a well-posed problem

LECTURE TERMINOLOGY (SEP 10) — HADAMARD WELL-POSEDNESS

A mathematical model is called **well posed** if

1. **a solution exists** (otherwise the model predicts nothing);
2. **the solution is unique** (otherwise the model cannot decide between futures);
3. **the solution depends continuously on the data** (a small error in the measured $x_0$ produces a small error in $x(t)$, at least over finite times).

The words "well-posed" and "Picard's theorem" are lecture terminology; Robinson never uses them. He discusses existence and uniqueness in Sec. 6.1 and states the result as Theorem 6.2 (lecture map, notes on differences).

Robinson gives the physical reading of uniqueness: if $t$ is time and the initial condition is specified "now", a unique solution means the equation can be used to *predict the future*; uniqueness is equivalent to the model being **deterministic** [p. 39]. It also legitimises guessing: if a guessed function satisfies the IVP, it is *the* solution.

### 7.2 Existence is not automatic

AN IVP WITH NO SOLUTION (ROBINSON P. 38)

$$x^2 + t^2 \frac{dx}{dt} = 0, \qquad x(0) = c, \quad c \neq 0.$$

Evaluate the equation at $t = 0$: the second term vanishes and $x(0)^2 = 0$, i.e. $x(0) = 0$. This contradicts $x(0) = c \neq 0$, so no differentiable solution exists. The culprit is visible in the explicit form $\dot{x} = -x^2/t^2$: the right-hand side is not even defined at $t = 0$.

SECTION 08 • ROBINSON SEC. 6.2, EX. 6.2–6.4 • LECTURE SEP 10

## 8. Picard's existence and uniqueness theorem

THEOREM 6.2 (PICARD'S THEOREM IN THE LECTURE) [ROBINSON, P. 40]

If $f(x,t)$ and $\dfrac{\partial f}{\partial x}(x,t)$ are continuous for $a < x < b$ and $c < t < d$, then for any $x_0 \in (a,b)$ and $t_0 \in (c,d)$ the initial value problem

$$\frac{dx}{dt} = f(x,t), \qquad x(t_0) = x_0$$

has a unique solution on some open interval $I$ containing $t_0$.

Three features deserve attention:

- **The hypothesis is easy to check:** continuity of $f$ and of one partial derivative, in a rectangle around $(x_0, t_0)$.
- **The derivative is with respect to the unknown** $x$, not $t$. Nothing is required of $\partial f/\partial t$.

- **The conclusion is local:** it says nothing about how large $I$ is (Robinson Fig. 6.2). Section 9 shows that $I$ can indeed be short.

## 8.1 The Lipschitz condition

**DEFINITION — LIPSCHITZ CONTINUITY** [ROBINSON, FOOTNOTE TO THM 6.2, EQ. (6.5), P. 40]

$f$ is **Lipschitz continuous in** $x$ if there is a constant $L$ such that

$$|f(x,t) - f(y,t)| \le L\,|x - y| \tag{6.5}$$

for all $x, y$ in the region considered. Robinson notes that this weaker hypothesis already suffices for Theorem 6.2.

**PROPOSITION — A BOUNDED DERIVATIVE IMPLIES LIPSCHITZ** [ROBINSON, EXERCISE 6.2, P. 44]

If $f$ is differentiable in $x$ with $\left|\dfrac{\partial f}{\partial x}\right| \le L$ for $a \le x \le b$, then $|f(x,t) - f(y,t)| \le L|x-y|$ there.

**PROOF (MEAN VALUE THEOREM)**

Fix $t$ and $a \le y < x \le b$. By the Mean Value Theorem there is $\xi \in (y,x)$ with $f(x,t) - f(y,t) = (x-y)\,\dfrac{\partial f}{\partial x}(\xi,t)$. Taking absolute values and using the bound, $|f(x,t)-f(y,t)| = |x-y|\,\left|\dfrac{\partial f}{\partial x}(\xi,t)\right| \le L\,|x-y|$. The case $x = y$ is trivial, and $x < y$ follows by symmetry.

∎

A continuous function on a closed bounded rectangle is bounded, so the hypothesis of Theorem 6.2 gives $|\partial f/\partial x| \le L$ on any closed sub-rectangle, and hence a Lipschitz bound. The converse is false: $f(x) = |x|$ is Lipschitz with $L = 1$ (because $\big||x| - |y|\big| \le |x-y|$) but not differentiable at $0$.

## 8.2 Proof of uniqueness – and of continuous dependence

Robinson leaves the proofs as guided exercises. The uniqueness argument (Exercise 6.3) is short, and the same estimate also delivers the third Hadamard condition.

**PROPOSITION — STABILITY ESTIMATE**

Suppose $|f(x,t) - f(y,t)| \le L|x-y|$ and let $x(t), y(t)$ solve $\dot{x} = f(x,t)$ with $x(t_0) = x_0$, $y(t_0) = y_0$. Then for $t \ge t_0$

$$|x(t) - y(t)| \le |x_0 - y_0|\, e^{L(t - t_0)}.$$

In particular, if $x_0 = y_0$ then $x(t) = y(t)$: **the solution is unique**.

PROOF (ROBINSON EXERCISES 6.3 AND 9.7)

1. Let $z(t) = x(t) - y(t)$. Subtracting the two equations: $\dot z = f(x(t),t) - f(y(t),t)$.
2. Let $Z(t) = z(t)^2 = |z|^2$. By the chain rule with outer function $u \mapsto u^2$:

$$\frac{dZ}{dt} = 2z\,\frac{dz}{dt} = 2z\,\big[f(x,t) - f(y,t)\big].$$

3. Bound it with the Lipschitz condition: $\dfrac{dZ}{dt} \le 2|z|\,|f(x,t) - f(y,t)| \le 2|z| \cdot L|z| = 2L\,Z.$
4. Multiply by the positive factor $e^{-2Lt}$ and use the product rule:

$$\frac{d}{dt}\Big(e^{-2Lt}Z\Big) = e^{-2Lt}\dot Z - 2Le^{-2Lt}Z = e^{-2Lt}\big(\dot Z - 2LZ\big) \le 0.$$

5. A function with non-positive derivative is non-increasing, so for $t \ge t_0$: $e^{-2Lt}Z(t) \le e^{-2Lt_0}Z(t_0)$, i.e. $Z(t) \le Z(t_0)\,e^{2L(t-t_0)}$. Taking square roots: $|z(t)| \le |z(t_0)|\,e^{L(t-t_0)}$.
6. If $x_0 = y_0$ then $z(t_0) = 0$, so $|z(t)| \le 0$ and $x(t) = y(t)$ for all $t \ge t_0$.

■

**Interpretation.** Two solutions that start $\varepsilon$ apart remain within $\varepsilon e^{L(t-t_0)}$ of each other: the error can grow, but at most exponentially, and it tends to zero with $\varepsilon$ on any finite time interval. This is exactly continuous dependence on the initial data. For $\dot x = \lambda x$ with $\lambda \ge 0$ (so $L = \lambda$) the bound is attained: $|x(t) - y(t)| = |x_0 - y_0|e^{\lambda(t-t_0)}$; for $\lambda < 0$ the solutions actually approach each other.

## 8.3 Idea of the existence proof: Picard iteration

Existence is harder than uniqueness. Robinson's Exercise 6.4 sketches it for $\dot x = f(x), x(0) = x_0$ [pp. 44–45].

1. **Integral form.** Integrating the ODE from 0 to $t$ (Barrow: $\int_0^t \dot x\,d\tilde t = x(t) - x(0)$) gives the equivalent integral equation

$$x(t) = x_0 + \int_0^t f\big(x(\tilde t)\big)\,d\tilde t. \tag{E6.5}$$

   Conversely, differentiating (E6.5) with the Fundamental Theorem of Calculus returns $\dot x = f(x)$, and $t = 0$ returns $x(0) = x_0$.

2 **Iteration.** Start from the constant guess $x_0(t) = x_0$ and improve it repeatedly:

$$x_{n+1}(t) = x_0 + \int_0^t f\big(x_n(\tilde{t})\big)\, d\tilde{t}. \tag{E6.6}$$

3 **Contraction.** Using (E6.4) $|f(x) - f(y)| \le L|x - y|$, for $0 \le t \le 1/(2L)$:

$$|x_{n+1}(t) - x_n(t)| \le \int_0^t L\,|x_n - x_{n-1}|\, d\tilde{t} \le L\,t\,\max|x_n - x_{n-1}| \le \tfrac{1}{2}\max|x_n - x_{n-1}|. \tag{E6.7}$$

Each step halves the maximum difference, so the differences shrink like $2^{-n}$ and $x_n(t)$ converges to some $x_\infty(t)$.

4 **Limit.** Letting $n \to \infty$ in (E6.6) shows that $x_\infty$ satisfies (E6.5), so it solves the IVP; the previous proposition shows it is the only one.

**ILLUSTRATION: PICARD ITERATES FOR $\dot{x} = x$, $x(0) = 1$**

Here $f(x) = x$ and $L = 1$. Each iterate integrates the previous one:

$$x_1 = 1 + \int_0^t 1\, d\tilde{t} = 1 + t, \qquad x_2 = 1 + \int_0^t (1 + \tilde{t})\, d\tilde{t} = 1 + \Big[\tilde{t} + \tfrac{\tilde{t}^2}{2}\Big]_0^t = 1 + t + \frac{t^2}{2},$$

$$x_3 = 1 + \Big[\tilde{t} + \tfrac{\tilde{t}^2}{2} + \tfrac{\tilde{t}^3}{6}\Big]_0^t = 1 + t + \frac{t^2}{2} + \frac{t^3}{6}, \qquad \dots, \qquad x_n = \sum_{j=0}^{n} \frac{t^j}{j!} \;\longrightarrow\; e^t.$$

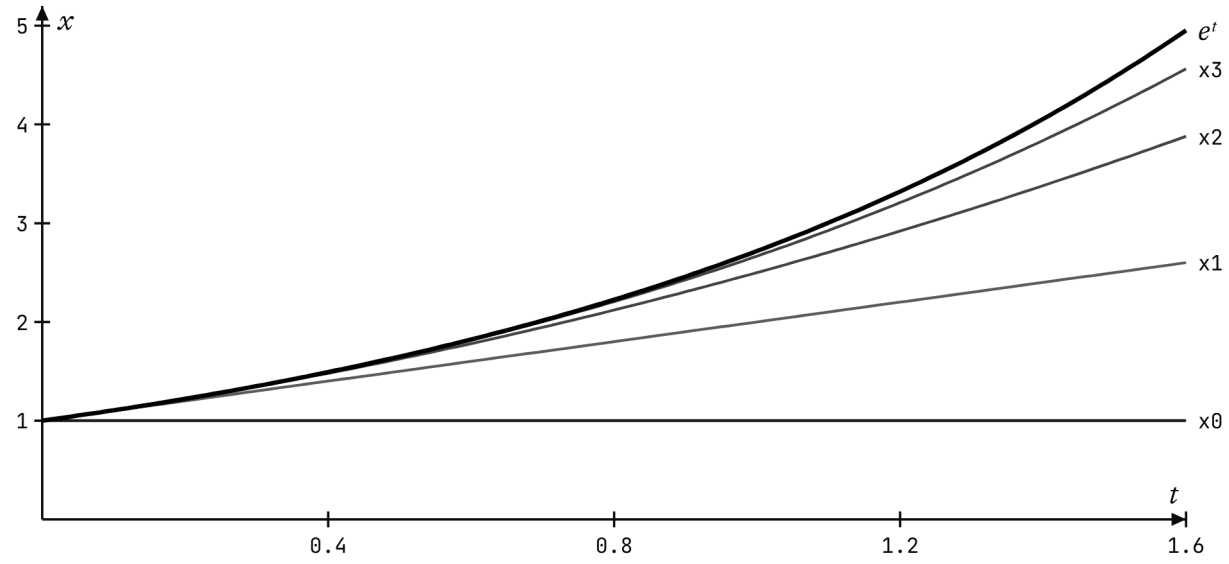



***Figure 1.7.*** *The first Picard iterates $x_0 = 1$, $x_1 = 1 + t$, $x_2 = 1 + t + t^2/2$, $x_3$ for $\dot{x} = x$, $x(0) = 1$, approaching the exact solution $e^t$ (indigo). Each iterate is the Taylor polynomial of $e^t$.*

SECTION 09 • ROBINSON SEC. 6.1, 6.3, 6.4, EX. 6.1 • LECTURE SEP 10

# 9. When the theorem fails or stops short

## 9.1 Loss of uniqueness: $\dot{x} = \sqrt{x}$

Robinson's example [Eq. (6.3), p. 39]:

$$\frac{dx}{dt} = \sqrt{x}, \qquad x(0) = 0. \tag{6.3}$$

The obvious solution is $x(t) \equiv 0$. But for any $c > 0$ the function

$$x_c(t) = \begin{cases} 0, & t \le c, \\ \dfrac{(t-c)^2}{4}, & t > c \end{cases}$$

is also a solution. **Check** for $t > c$ (chain rule with inner function $t - c$): $\dot{x}_c = \dfrac{2(t-c)}{4} \cdot 1 = \dfrac{t-c}{2}$, while $\sqrt{x_c} = \sqrt{(t-c)^2/4} = \dfrac{|t-c|}{2} = \dfrac{t-c}{2}$. At $t = c$ both one-sided derivatives are $0$, so $x_c$ is differentiable there. The solution "waits" at $0$ and then "decides" at the arbitrary time $c$ to leave [p. 39]: the model is not deterministic.

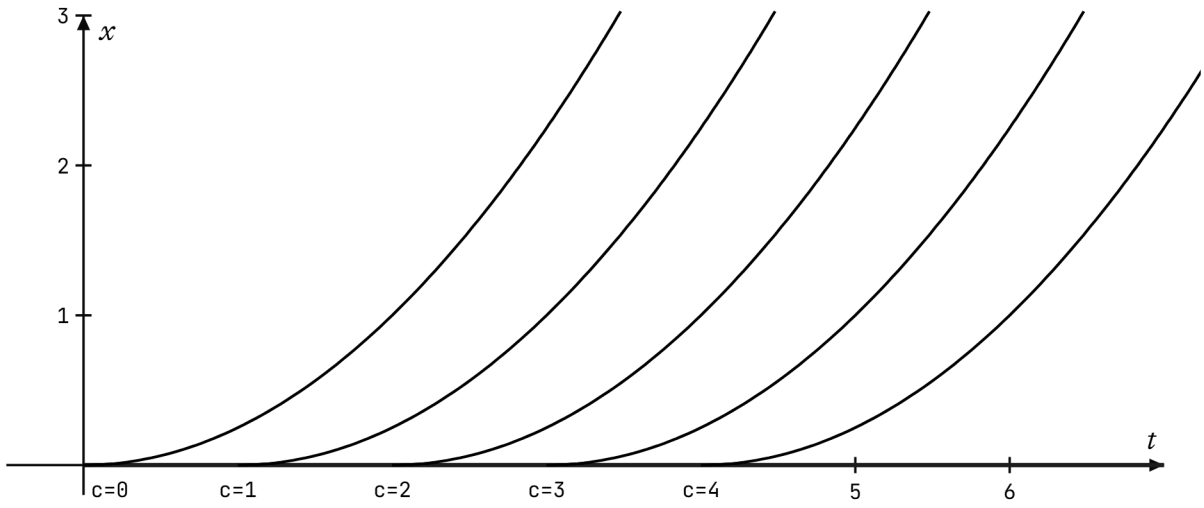



***Figure 1.5.*** *Infinitely many solutions of $\dot{x} = \sqrt{x}$, $x(0) = 0$ (Robinson (6.3), Fig. 6.1, p. 39): the zero solution (red) and the curves $x_c(t)$ that stay at $0$ until $t = c$ and then follow $(t-c)^2/4$.*

**Why Theorem 6.2 does not apply.** $f(x) = x^{1/2}$ is continuous, but $f'(x) = \tfrac{1}{2}x^{-1/2} \to \infty$ as $x \downarrow 0$, so $\partial f/\partial x$ is not continuous at the initial point $x_0 = 0$ [p. 40; Fig. 6.3, p. 41]. Nor is $f$ Lipschitz near $0$: $\dfrac{|f(x)-f(0)|}{|x-0|} = \dfrac{\sqrt{x}}{x} = \dfrac{1}{\sqrt{x}} \to \infty$ as $x \downarrow 0$, so no constant $L$ works. Note the logic: the theorem gives a *sufficient* condition; its failure here is consistent with the non-uniqueness, not a proof of it – the explicit family $x_c$ is the proof.

**THE SAME PHENOMENON FOR $\dot{x} = x^{1/3}$ (ROBINSON EXERCISE 6.1 (III))**

With $x(0) = 0$, $x \equiv 0$ is a solution. Separating the variables for $x > 0$: $x^{-1/3}dx = dt$. An antiderivative of $x^{-1/3}$ is $\tfrac{3}{2}x^{2/3}$, so integrating from the time $c$ at which $x$ leaves $0$:

$$\left[\tfrac{3}{2}x^{2/3}\right]_0^{x(t)} = \left[\tilde{t}\right]_c^t \implies \tfrac{3}{2}x(t)^{2/3} = t - c \implies x(t) = \left(\frac{2(t-c)}{3}\right)^{3/2}, \quad t \geq c.$$

Check: $\dot{x} = \tfrac{3}{2}\left(\tfrac{2(t-c)}{3}\right)^{1/2} \cdot \tfrac{2}{3} = \left(\tfrac{2(t-c)}{3}\right)^{1/2} = x^{1/3}$. Again infinitely many solutions (and their mirror images $-x$), because $f'(x) = \tfrac{1}{3}x^{-2/3}$ blows up at $x = 0$.

## 9.2 Finite-time blow-up: $\dot{x} = x^2$

Robinson's second cautionary example [Sec. 6.3, Eq. (6.6), p. 41]:

$$\frac{dx}{dt} = x^2, \qquad x(0) = x_0. \tag{6.6}$$

Here $f(x) = x^2$ and $f'(x) = 2x$ are continuous everywhere, so Theorem 6.2 guarantees a unique solution near $t = 0$ for every $x_0$. For $x_0 \neq 0$ the solution is [Eq. (6.7), p. 42]

$$x(t) = \frac{1}{x_0^{-1} - t}. \tag{6.7}$$

**Verification.** At $t = 0$, $x = 1/x_0^{-1} = x_0$ . Write $x = u^{-1}$ with $u(t) = x_0^{-1} - t$, $du/dt = -1$. Chain rule:

$$\frac{dx}{dt} = \frac{d}{du}\big(u^{-1}\big) \cdot \frac{du}{dt} = \big(-u^{-2}\big) \cdot (-1) = \frac{1}{(x_0^{-1} - t)^2} = x(t)^2 \ \checkmark.$$

By uniqueness this is *the* solution. (Topic 2 derives it by separation of variables, Robinson Example 8.1.)

- $x_0 > 0$**:** the denominator $x_0^{-1} - t$ decreases to 0 at $t^* = 1/x_0$, so $x(t) \to +\infty$ as $t \uparrow 1/x_0$: the solution **blows up in finite time**. Backwards in time it decays to 0. Maximal interval of existence: $(-\infty, 1/x_0)$.
- $x_0 < 0$**:** the blow-up happens in the past, $x \to -\infty$ as $t \downarrow 1/x_0 < 0$. Maximal interval: $(1/x_0, +\infty)$.
- $x_0 = 0$**:** $x \equiv 0$, defined for all $t \in \mathbb{R}$.

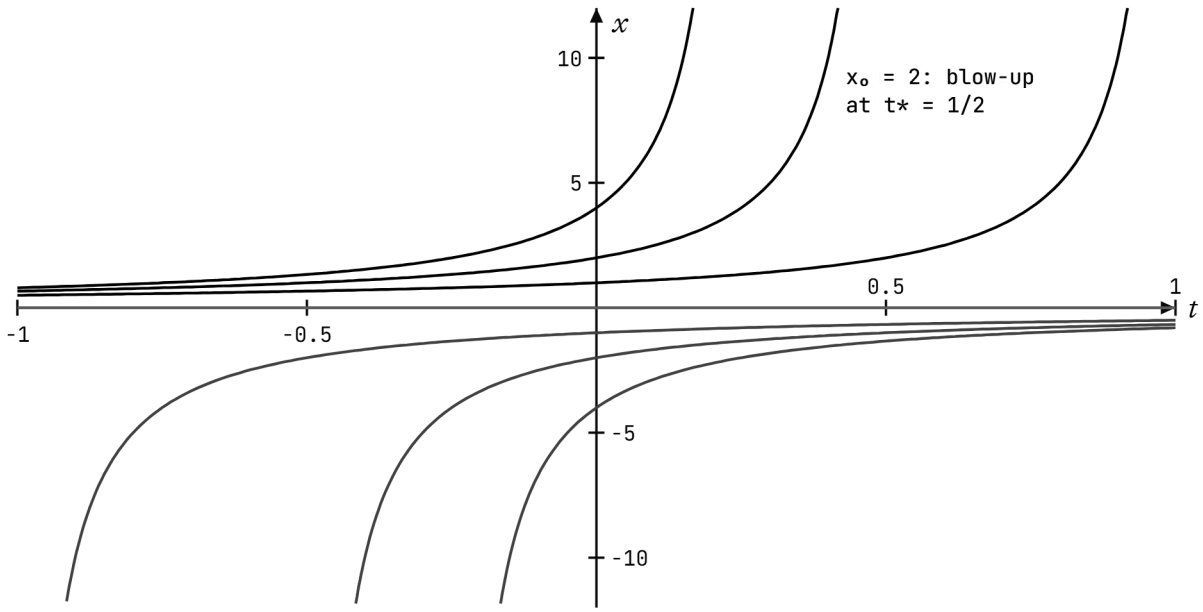



***Figure 1.6.*** *Solutions $x(t) = 1/(x_0^{-1} - t)$ of $\dot{x} = x^2$ for $x_0 = 1, 2, 4$ (indigo) and $x_0 = -1, -2, -4$ (green), cf. Robinson Fig. 6.4, p. 43. Positive data blow up at $t^* = 1/x_0 > 0$; negative data blow up in the past at $t^* = 1/x_0 < 0$; only $x_0 = 0$ gives a solution on all of $\mathbb{R}$.*

WHY THE INTERVAL IN THEOREM 6.2 MUST BE "SOME INTERVAL"

$f(x) = x^2$ is as smooth as a function can be, yet solutions do not exist for all time. The reason is that $f$ is Lipschitz on every bounded interval but not on all of $\mathbb{R}$: $|x^2 - y^2| = |x + y|\,|x - y|$ and $|x + y|$ is unbounded. Fast growth ($\dot{x} \sim x^2$) feeds on itself. Compare $\dot{x} = |x|$, whose solutions $x_0 e^t$ (for $x_0 > 0$) exist forever although its phase line looks identical (Robinson Sec. 7.5.3).

### 9.3 Summary of the failure modes

| EQUATION | HYPOTHESES OF THM 6.2 | WHAT HAPPENS | LESSON |
|---|---|---|---|
| $x^2 + t^2\dot{x} = 0,\ x(0) = c \neq 0$ | $f = -x^2/t^2$ undefined at $t = 0$ | No solution | Existence can fail |
| $\dot{x} = \sqrt{x},\ x(0) = 0$ | $\partial f/\partial x$ infinite at $x = 0$ | Infinitely many solutions | Uniqueness needs Lipschitz |
| $\dot{x} = x^{1/3},\ x(0) = 0$ | $\partial f/\partial x$ infinite at $x = 0$ | Infinitely many solutions | Same as above |
| $\dot{x} = x^2,\ x(0) = x_0 > 0$ | Satisfied everywhere | Unique, but blows up at $t = 1/x_0$ | The theorem is only local |

A ONE-MILLION-DOLLAR OPEN PROBLEM (ROBINSON SEC. 6.4, PP. 42–43)

It is still not known whether the three-dimensional Navier–Stokes equations have unique solutions that exist for all positive times. Every CFD computation of a wing or an engine assumes they do, yet one cannot prove that a numerical approximation is close to the "true solution" without knowing that such a solution exists. The question is one of the Clay Mathematics Institute's seven Millennium Prize Problems: prove global existence and uniqueness, or find one initial condition for which the solution breaks down.

SECTION 10 • SUMMARY

## 10. Summary and lecture map

| IDEA | KEY STATEMENT | CLASS NOTES | ROBINSON | LECTURE |
|---|---|---|---|---|
| ODE vs PDE | One independent variable vs several; $x' = x$, heat equation | p. 2 | Sec. 3.1, p. 11 | Sep 8 |
| Order, autonomy | Highest derivative; no explicit $t$ | p. 2 (autonomy p. 3) | Sec. 3.2, p. 13 | Sep 8 |
| Linearity | Form (L); superposition test: $u' = u$ linear, $u'' = (u')^2 + 1$ not | p. 2 | Sec. 3.3, p. 13–14 | Sep 8–10 (corrected) |

| IDEA | KEY STATEMENT | CLASS NOTES | ROBINSON | LECTURE |
|---|---|---|---|---|
| Systems | Two or more dependent variables; $x' = 2x - y$, $y' = -x + 3y$ | p. 2 | Eq. (3.9), p. 15; Ch. 25 | Sep 8 |
| Population growth / decay | $x' = kx \Rightarrow x = x_0 e^{kt}$; grows if $k > 0$, decays if $k < 0$ | p. 2 | Sec. 8.2–8.3, p. 61–64 | Sep 8 |
| Logistic correction | $x' = kx(1 - x/M)$, $x \to M$; solution in 5.4 | p. 2 (analysis p. 3) | Sec. 8.5, p. 66–68 | Sep 8–10 |
| Free fall | $-mg = my'' \Rightarrow y = -\frac{1}{2}gt^2$ (from rest at 0) | p. 2 | Sec. 5.3, Ex. 5.6 | Sep 8 |
| Quadratic drag | $v = w_T \tanh(gt/w_T)$, $w_T = \sqrt{mg/k}$ | – | Sec. 7.5.2, Ex. 8.6 | – |
| IVP, well-posedness | Existence, uniqueness, continuous dependence | p. 3 | Sec. 6.1, p. 38 | Sep 10 |
| Picard (Thm 6.2) | $f$, $\partial f/\partial x$ continuous ⇒ unique local solution | p. 3 | Sec. 6.2, p. 40 | Sep 10 |
| Failure modes | $\sqrt{x}$: non-uniqueness; $x^2$: blow-up | – | Sec. 6.1, 6.3, p. 39–42 | Sep 10 |

**WHAT COMES NEXT**

Topic 2 develops the methods announced here: the phase line for autonomous equations (Ch. 7), separation of variables with its justification (Ch. 8), the integrating factor (Ch. 9) and exact equations (Ch. 10).